Prototyping Interactions Course Syllabus

Corcoran School of the Arts & Design | The George Washington University

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Course and Contact Information:

Course: Prototyping Interactions

Semester: Fall 2019
Meeting time: Thursdays
8:00am-12:00
Location: Physical Computing Lab

Instructor:

Name: Chloe Varelidi
Campus : Corcoran School of the Arts & Design
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Course description:

Prototyping Interactions is about making ideas tangible -not perfect-, learning through making and quickly getting key feedback from the people you are designing for; it is the how behind the way people and organizations develop products, services, strategy, and impact. Prototyping is something you can learn only by doing, so we ll tinker with multiple prototyping tools (laser cutting, 3D printing, hardware and software prototyping) through short weekly design challenges for the first half of the semester, with a longer project for the final part. At the end of the class you should feel confident in your abilities to quickly prototype and iterate solutions to any design problem you are faced with and come up with new ideas as a result.

Course prerequisites, if any:

There are no prerequisites for this course. However it is encouraged that students have some knowledge of vector design programs like Adobe Illustrator and 3D design programs like Solidworks.

Learning outcomes:

Upon successful completion of this course, students should be able to:

1. Apply different digital and non digital fabrication techniques to create interactive prototypes.
2. Create rapid prototypes and user test them.
3. Collaborate effectively with peers and synthesize constructive feedback.
4. Analyze user feedback to iterate on their prototypes to improve them.
5. Take on mindsets such as “learning from failure”, “empathy” and “iteration” to create innovative solutions.
6. Apply the prototyping cycle of create-test-iterate to solve real world problems.

7. Create user journey maps and storyboards for their prototypes to communicate how their ideas will work in a real life.

Required textbooks, materials, and recommended readings:

Readings:

All readings are strongly recommended but not mandatory. A selection of shorter weekly readings from these books will be provided in digital format during the semester.

1. Erika Hall: *Just Enough Research*
2. *Macklin C. and Sharp J. (2016) Games, Design & Play, Pearson*
3. *Kristina Hook: Designing With The Body, Somaesthetic Interaction Design, MIT Press*

Materials:

Notebook with blank pages and pen. Mandatory, please have a dedicated notebook for the class.

Weekly assignments: materials will be provided by the University and will be available on loan, they will be logged and need to be returned at the end of the assignment.

Midterms & Finals: You will need to purchase materials for your midterm and final projects.

Average minimum amount of out-of-class or independent learning expected per week:

This course requires a minimum of 7.5 hours of out-of-class work per week.

Schedule:

Date	Topic(s) and readings	Assignment(s) Due
WEEK 1 08/29	Introduction to Prototyping Interactions. Lasercutter Basics.	No assignment due this week. NOTE: Find a cheap electronic toy to bring in class for next week.
WEEK 2 09/05	Toy Hacking. User Testing Guides. Reading: Erika Hall	Lasercut Assignment Due
WEEK 3 09/12	Conductive Materials. Soft Circuits.	Toy Assignment Due
WEEK 4 09/19	Circuits & Code I: Inputs, Outputs, Loops, Logic, Variables.	Soft Circuit Assignment Due

WEEK 5 09/26	Circuits & Code II: Functions Game Design Prototyping Reading: *Macklin C. and Sharp J.*	Circuit Assignment Due
WEEK 6 10/03	3D Printing I	Game Assignment Due
WEEK 7 10/10	Midterm Project: Ideas, Prototyping	3D Printing Assignment Due
WEEK 8 10/17	Midterms: In class prototype development	Midterm Project Work In Progress Presentation Due
WEEK 9 10/24	Midterm Presentations	Midterm Project & Presentation Due
WEEK 10 10/31	Web prototyping	No assignment due this week.
WEEK 11 11/07	Final Project	Internet Meme Assignment Due Final Project Themes Presentations
WEEK 12 11/14	Final Project: Empathy 360 Maps, Defining the problem, value proposition	Group Presentations: Review Work in Progress
WEEK 13 11/21	Final Project: Prototyping multiple concepts, in class user testing.	Group Presentations: Review Work in Progress
WEEK 14 12/05	Final Project: Prototyping multiple concepts, in class user testing.	Group Presentations: Review Work in Progress
WEEK 15 12/12	Finals	Final Project Presentations

Assignments

Assignment	Description	Total Points
Lasercutting due 01/22	(1) Design and lasercut a simple boardgame like chess or backgammon using a material from fablab: i.e. cardboard and plywood.	5
Toy Hacking 01/29	(1) Complete in class activity of taking apart a cheap toy and creating a new case to house its mechanism. (2) Share your prototype and learnings in class.	5

Soft Circuits due 02/05	(1) Complete in class activity of building your notebook soft circuit. (2) Create one iteration of your circuit (shape, placement or number of LEDs, conductive material). (3) Share your prototypes and	5
Circuits & Code I due 02/12	(1) Complete in class mini activities of building and programming different littleBit circuits. (2) Create at least 3 iterations of your circuits. (3) Share your prototypes and	5
Circuits & Code II due 02/19	(1) Complete in class activity of building and coding a game using littleBits. (2) Share your prototype and learnings	5
3D Printing due 02/26	(1) Complete in class activity of 3D printing a small object. (2) Optional: design and 3D print a miniature object of a simple found object. (3) Share your prototype and learnings	5
Midterms due 03/05	(1) Complete midterm project to create a robot using any prototyping technique of your choice. (2) Share your prototype and learnings in class. NOTE: Midterm presentation requirements will be outlined beforehand.	25
Web Prototyping 03/ 26	(1) Create an internet meme using HTML & CSS. (2) Final Project Ideas Presententation	5
Finals	(1) Weekly group presentations of work in progress. (2) Final project: real world design challenge. Guidelines provided. (3) Final presentations with industry experts.	40
	Total Possible Points	100

Grading

- weekly design challenges (35%)
- midterm project (25%)
- final project (40%)

NOTE: in class participation and team work calculated is taken into account in the grade breakdown above.

About the Instructor

Chloe Varelidi is a product designer, educator and the founder of the design studio Humans Who Play. She is a graduate of the Design & Technology MFA program at Parsons, The New School for Design as well as an Eyebeam Art & Technology Resident Alumna. Prior to starting her own practice she was part of the early teams at the Institute of Play, Quest to Learn, the Mozilla Foundation and littleBits launching everything from top rated apps, to franchise products for Marvel & Disney, to a public school featured on the cover of the NYT Magazine and winning multiple awards for her work along the way, including some from ISTE, TOTY and Common Sense Media. She is recognized as one of the GOOD100 for shaping the world in meaningful and creative ways and frequently lectures about harnessing play as a force for innovation and doing good.

Course Policies

All assignments are due at the beginning of the class. Assignments are typically shared in class, so failure to complete your assignments effects the entire class.

Students should always keep a backup copy of their work.

Late Assignments:

No late assignments will be accepted. In this course, assignments build on the previous. Failure to complete prior assignments will make each subsequent assignment more difficult. It is in your best interest to complete each assignment on time and to the best of your ability. Always hand in what you have, even if it does not work. Partial credit is better than no credit at all.

Attendance / Absences:

Students are expected to attend each class and arrive on time. Any student arriving late for an exam or quiz may not be given a chance to complete it.

Late assignments are not accepted unless they result from an excused absence. Excused absences are limited to documented medical emergencies and events for which the instructor has given approval. All students are expected to communicate planned or unplanned absence to the instructor's email as soon as possible.

Any student accruing more than a 20% unexcused absence rate will receive a full grade deduction. If, for example, a class meets 10 times during a semester, the student's third absence will result in a best potential grade of "B." A student who accrues 30% or more unexcused absences will fail the course.

Makeup exams and acceptance of late assignments will only be granted in the following circumstances; Medical excuse, emergencies (as understood by GW University Administration), campus-sponsored activities.

All planned absences should be clearly explained in an email sent to the instructor before the student misses the class. The instructor will reply indicating whether or not the absence is excused.

All issues of attendance and tardiness will be handled as school policy dictates and at the discretion of the instructor.

Correspondence:

All students are expected to check their GWU supplied email daily, or forward email to an account they do check daily.

In Class Conduct:

In-class web surfing, email, electronic chat, text messaging, or related behavior is prohibited during class meetings unless specifically requested by the instructor.

Please be attentive to people's comments and engage yourself in class.

University policies and resources for students:

University policy on observance of religious holidays

In accordance with University policy, students should notify faculty during the first week of the semester of their intention to be absent from class on their day(s) of religious observance. For details and policy, see: students.gwu.edu/accommodations-religious-holidays.

Academic integrity code

Academic dishonesty is defined as cheating of any kind, including misrepresenting one's own work, taking credit for the work of others without crediting them and without appropriate authorization, and the fabrication of information. For details and complete code, see: studentconduct.gwu.edu/code-academic-integrity

Support for students outside the classroom

Disability Support Services (DSS)

Any student who may need an accommodation based on the potential impact of a disability should contact the Disability Support Services office at 202-994-8250 in the Rome Hall, Suite 102, to establish eligibility and to coordinate reasonable accommodations. For additional information see: disabilitysupport.gwu.edu

Counseling and psychological services 202-994-5300

GW's Colonial Health Center offers counseling and psychological services, supporting mental health and personal development by collaborating directly with students to overcome challenges and difficulties that may interfere with academic, emotional, and personal success. For additional information see healthcenter.gwu.edu/counseling-and-psychological-services.

Safety and security

In case of an emergency, if at all possible, the class should shelter in place. If your building is affected, follow the evacuation procedures and seek shelter at a predetermined rendezvous location. GW Alert is the university's notification system that sends emergency text message and email alerts to the GW community. Students are requested to maintain current contact information by logging on to alert.gwu.edu. Download the GW Personal Alarm Locator (GW PAL), a mobile safety and security application that allows users to alert GWPD of a crime, report crime tips anonymously, provide a safety profile, and identify their location in real time. For more safety and security information and tips, visit safety.gwu.edu.